Abdelrahman Mohamed Alnaqeeb

abdelrahmanalnaqeeb@gmail.com | naqeeb1124.github.io/abdelrahman-portfolio | github.com/Naqeeb1124

Research interests: PDE-constrained and topology optimization, adjoint methods, optimal control, computational fluid dynamics

Education

Zewail City of Science and Technology Expected June 2027
B.Sc. in Aerospace Engineering *GPA: 3.3/4.0 (to date)*
Relevant coursework: Numerical Analysis; Probability & Stochastic Processes; Fluid Mechanics II; Gas Dynamics; Control Systems; Advanced Control; Flight Dynamics & Control; Jet Propulsion; Machine Design II.

Industry Research Internship

Lotus Power — CFD and Machine Learning Researcher Jun 2026 – present
Adjoint-based topology optimization of airflow in a 20-tray industrial dryer *Supervisor: Dr. Basem Badr*

- Posed tray-airflow uniformity as PDE-constrained topology optimization with a RAMP-interpolated Brinkman penalty.
- Formulated the 2D Navier–Stokes/Brinkman state model (staggered MAC grid, upwind) with nonlinear fan-curve actuators.
- Derived the Newton Jacobian including the fan-curve dP/dQ coupling; specified damped Newton with force continuation.
- Derived discrete-adjoint sensitivities through a density filter and tanh projection; designed finite-difference checks.
- Designed the optimality-criteria update: multiplier bisection, move limits, trust-region backtracking, best-iterate restore.
- Objective: tray-speed CV^2 plus a speed-band hinge, volume-band penalty and 700 Pa worst-bank fan-pressure cap.
- Ran penalization/projection continuation studies; diagnosed divergent and baffle-erasing schedules, adopted a stable one.
- Cut tray-speed CV 0.495 $\rightarrow$ 0.09 (best iterate) and peak fan pressure 1115 $\rightarrow$ 544 Pa (2D reduced model; not 3D-validated).
- Directed an AI-assisted Python implementation (SciPy sparse, pytest); vectorized assembly gave a 14.5$\times$ speed-up.
- Flow-budget analysis: the original fan bank (0.83 m^3/s free delivery) could not feed 20 trays at 3 m/s; drove fan re-selection.
- Set up a 3D CFD pipeline (build123d, PyFluent meshing, k–ω SST, 7.2M cells) for later iterations; not yet converged.
- Built a 1D hydraulic-network fan-sizing model on 53 digitized catalog fan curves, with loss-coefficient sensitivity sweeps.

Research Experience

Graduation Thesis Sep 2026 – Jun 2027 (expected)
Mean-Field-Game Guidance of a Fixed-Wing Responder Fleet for Swarm Containment
Advisor: Dr. Ahmed Abdelsamea · Major coordinator: Dr. Ahmed Eltaweel
Status: formulation and experiment design drafted (scope revision v2, Oct 2026, under advisor review); solvers and simulations scheduled Nov 2026 – Apr 2027

- Formulated a team-versus-crowd mean-field game: a coupled HJB–Fokker–Planck optimality system with five PDEs.
- Specified the planner solve by adjoint gradients with Armijo backtracking inside damped fictitious play.
- Specified exploitability and stationarity residuals to certify each equilibrium iterate, with continuation to map multiplicity.
- Retained a convex mean-field-control benchmark and an exact LQ Riccati reference for solver verification.
- Specified a fidelity ladder to a 6-DOF fixed-wing UAV with autopilot and a planned Latin-hypercube envelope study.

Selected Projects

Topology Optimization of a Satellite Bracket Sep 2025
Self-directed project; SolidWorks Simulation

- Linear-static FEA baseline (18k elements, 82,770 DOF); compliance minimization with 45% mass limit, stress cap.
- Converged in 37 iterations on a refined 71k-element mesh with frozen (non-design) regions.

Cold Plate Design and Channel Topology Optimization May 2025
Heat Transfer course project on the ASME K-16/IEEE EPS brief; led all technical work in a 3-person team; Python, COMSOL

- Derived adjoint/KKT conditions for density-based fluid–thermal channel optimization (Brinkman Navier–Stokes + energy).
- Screened 10 fin concepts by figure of merit; ran COMSOL channel topology optimization for two inlet layouts.

Ramjet Design Optimization Jan 2026
Course project, 2-person team; MATLAB

- Grid-searched two-ramp intake angles under shock-attachment and 2200 K limits: 6.26 kN net thrust at Mach 2.75.
- Off-design: predicted intake unstart at Mach 2.0; at Mach 4.0 thrust rose to the 2200 K limit (13.7 kN).

JUICE Interplanetary Trajectory Reconstruction Jan 2026
Individual project; MATLAB, SPICE

- Built a SPICE-anchored MGA-1DSM model with a universal-variable Lambert solver; optimized DSM timing with SQP.

Control and GNC Projects 2025 – 2026
Individual and team projects; MATLAB/Simulink

- CubeSat ADCS digital twin built via the Simulink API: B-dot detumbling, quaternion PD nadir tracking, wheel desaturation.
- Quarter-car active suspension (4-person team): built the discrete LQR Simulink model, terrain GUI and ride-comfort study.
- Boeing 747 attitude autopilots: compiled stability derivatives; tuned pitch (PID) and roll (PI) attitude-hold loops.

Skills

Mathematics	Linear algebra, ODEs/PDEs, numerical analysis, probability & stochastic processes, optimization
Methods	Discrete adjoints, density-based topology optimization (OC, filtering, projection), Newton continuation, sparse linear algebra; SQP; mean-field games (HJB–FPK); RANS CFD; state-space control, LQR
Software	Python (NumPy, SciPy sparse, JAX, pytest), MATLAB/Simulink, ANSYS Fluent/PyFluent, OpenFOAM, COMSOL, SolidWorks Simulation, build123d, Git, LaTeX